

Problem - Solution Fit

DATE	02 JULY 2026
TEAM ID	
PROJECT NAME	A Comprehensive Measure of Well-Being: Human Development Index (HDI) Predictor
MAXIMUM MARKS	2 MARKS

1. Overview

The Problem-Solution Fit document evaluates how the system's design and implemented features address the core problem statements identified during the ideation phase. This analysis ensures that every feature has a clear purpose and directly solves a user need.

2. Problem-Solution Mapping

Problem ID	Identified Problem Statement	Implemented Solution Feature	Fit Justification
P1	Delayed and retrospective official data reporting	Real-time predictive ML engine	Users can input immediate, localized, or hypothetical indicators and obtain an instant prediction, bypassing annual reporting delays.
P2	Complexity in calculating multi-dimensional composite indices	Automated Linear Regression	The Flask backend handles data preprocessing, encoding, and model execution automatically, abstracting math calculations away from the user.
P3	Inability to run "what-if" planning scenarios	Interactive Input Form (`/predict`)	Users can change individual sliders/numbers (e.g. adding 2 years to school expectancy)

			and view the immediate effect on the final score.
P4	Lack of actionable recommendations from raw indices	Contextual Recommendations Engine	The system maps the predicted tier to targeted recommendations (e.g., healthcare expansion, education reforms), providing clear next steps.
P5	Poor accessibility and technical presentation of data	Glassmorphic Progress Gauge	A responsive web interface presents results visually, using color-coded badges and progress bars to make data accessible to non-technical users.

3. Scenario-Based Model Validation

To ensure the model performs correctly under different conditions, the team validated the system against three test scenarios representing different development levels.

Scenario 1: Very High Human Development

- **Input Parameters:**
- Country: `Country\_A`
- Life Expectancy: `82.0 years`
- Expected Years of Schooling: `16.0 years`
- Mean Years of Schooling: `13.0 years`
- GNI per Capita: `\$55,000`
- **Model Output:** Predicted HDI = `1.0` (Clamped maximum).
- **Tier Assignment:** `Very High Human Development` ( green badge).
- **Insight Alignment:** Recommends advanced research investments and technology sector support.

Scenario 2: Medium/High Human Development

- **Input Parameters:**
- Country: `Country\_C`
- Life Expectancy: `62.0 years`
- Expected Years of Schooling: `10.0 years`
- Mean Years of Schooling: `6.0 years`

- GNI per Capita: `\$8,000`
- **Model Output:** Predicted HDI = `0.5842` to `0.7120` (Depending on country category).
- **Tier Assignment:** `Medium/High Human Development` ( /  orange/blue badge).
- **Insight Alignment:** Recommends secondary school enrollment expansions and manufacturing growth.

Scenario 3: Low Human Development

- **Input Parameters:**
- Country: `Country\_C`
- Life Expectancy: `52.0 years`
- Expected Years of Schooling: `5.0 years`
- Mean Years of Schooling: `3.0 years`
- GNI per Capita: `\$1,500`
- **Model Output:** Predicted HDI = `0.5408` (Below 0.550 threshold).
- **Tier Assignment:** `Low Human Development` ( red badge).
- **Insight Alignment:** Recommends basic vaccination coverage, primary education funding, and humanitarian aid coordination.

4. Value Proposition

The HDI Prediction System replaces slow, retrospective, and manual calculations with a **one-click, interactive predictive dashboard**. It translates abstract statistical indices into clear visual formats and actionable policy recommendations, empowering stakeholders to plan development programs effectively.